Dear Editor,

We are pleased to submit our manuscript entitled *"PREDICT-AML: Personalized REsponse via Drug Interaction and Cellular Traits for Acute Myeloid Leukemia"* for your consideration. This study presents a comprehensive machine learning framework designed to forecast personalized drug responses in AML patients using the richly annotated BeatAML cohort.

PREDICT-AML addresses the key limitations of prior retrospective studies by introducing a robust, predictive, and interpretable model that integrates multi-omic features with pharmacological representations. We believe this work will be of strong interest to readers in the fields of computational oncology, machine learning, and precision medicine.

Key novelties of this study include:

- **Multi-modal patient feature engineering** incorporates gene expression, mutations, pathway activity, immune cell states, and clinical annotations to comprehensively model drug sensitivity.
- **Dual drug representation strategies** using both Morgan Fingerprints and deep SMILES embeddings (via TF-LSTM) to capture chemical and structural nuances of compounds.
- **Integration of patient-specific drug-pathway proximity** via network propagation on protein interaction networks, enabling mechanistic alignment between drugs and oncogenic pathways.
- **Systematic benchmarking** of traditional and deep learning models, with CatBoost emerging as the optimal predictor (test MAE: 39.04; Pearson's *r*: 0.66), showing strong generalization on an unseen cohort.
- **Biological interpretability using SHAP** to identify predictive molecular features and pathway associations, supporting mechanistic insights and potential biomarker discovery.
- **Feature association analysis** reveals key modulators of drug sensitivity, including *CD300E*, *LGALS2*, and the immunogenic cell death (ICD) pathway, offering translational relevance.
- **Public release of code and workflows**, ensuring reproducibility and accessibility for the broader research community at https://github.com/raghvendra5688/PT-AML

We believe this work represents a significant advance toward data-driven, patient-specific therapeutic strategies in AML and lays the groundwork for future clinical translation. Thank you for considering our manuscript. We look forward to your feedback.

Our contact information are e-mail: hbensmail@hbku.edu.qa and raghvendramall@ieee.org . None of these data have been published or are under consideration elsewhere. Please don’t hesitate to contact us if you would like any additional information or have any questions.

With Regards,

Dr. Halima Bensmail and Dr. Raghvendra Mall

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